PerformanceTesting

Date	3July2026
TeamID	XXXXXX
ProjectName	EduGenie–AI–PoweredEducationalAssistant
MaximumMarks	5Marks

Step1:TestingOverview

Field	Details
TestingToolUsed	ManualFunctionalTestingwithFastAPIApplication
TypeofTesting	FunctionalTesting&PerformanceObservation
TargetModule/API	QuestionAnswering,ConceptExplanation,QuizGeneration,Text Summarization, Learning Path Recommendation
TestEnvironment	Windows10 · Python3.14 · FastAPI · HTML/CSS · Gemini API
TestDate	29 June 2026

Step2:TestScenarios

S.No	Scenario	Users	Duration	ExpectedOutcome
1	LaunchEduGenie Application	1	3–5 sec	Homepageloads successfully
2	AskEducational Question	1	4–8 sec	AIreturnsanaccurateanswerwitheducation context

3	GenerateConcept Explanation	1	5-8 sec	Simplifiedexplanationisdisplayed correctly
4	GenerateAI Quiz	1	5-10 sec	Topic-basedMCQsaregenerated successfully
5	SummarizeEducational Text	1	4-8 sec	Concisesummaryis generated
6	GenerateLearning Path	1	6-10 sec	Personalizedlearningroadmapis displayed
7	NavigateBetween Modules	1	Lessthan2 sec	Smoothpagetransitionswithout errors

Step3:Observations

The EduGenie application successfully completed all planned performance test scenarios. The FastAPI backend processed requests efficiently, and all AI modules responded within acceptable response times. Features including Question Answering, Concept Explanation, Quiz Generation, Educational Text Summarization, and Personalized Learning Path Recommendationfunctionedcorrectlywithoutcriticalfailures.Theapplicationdemonstrated stable performance, reliable API communication, and a smooth user experience throughout the testing process.

Step4:Screenshots/Evidence

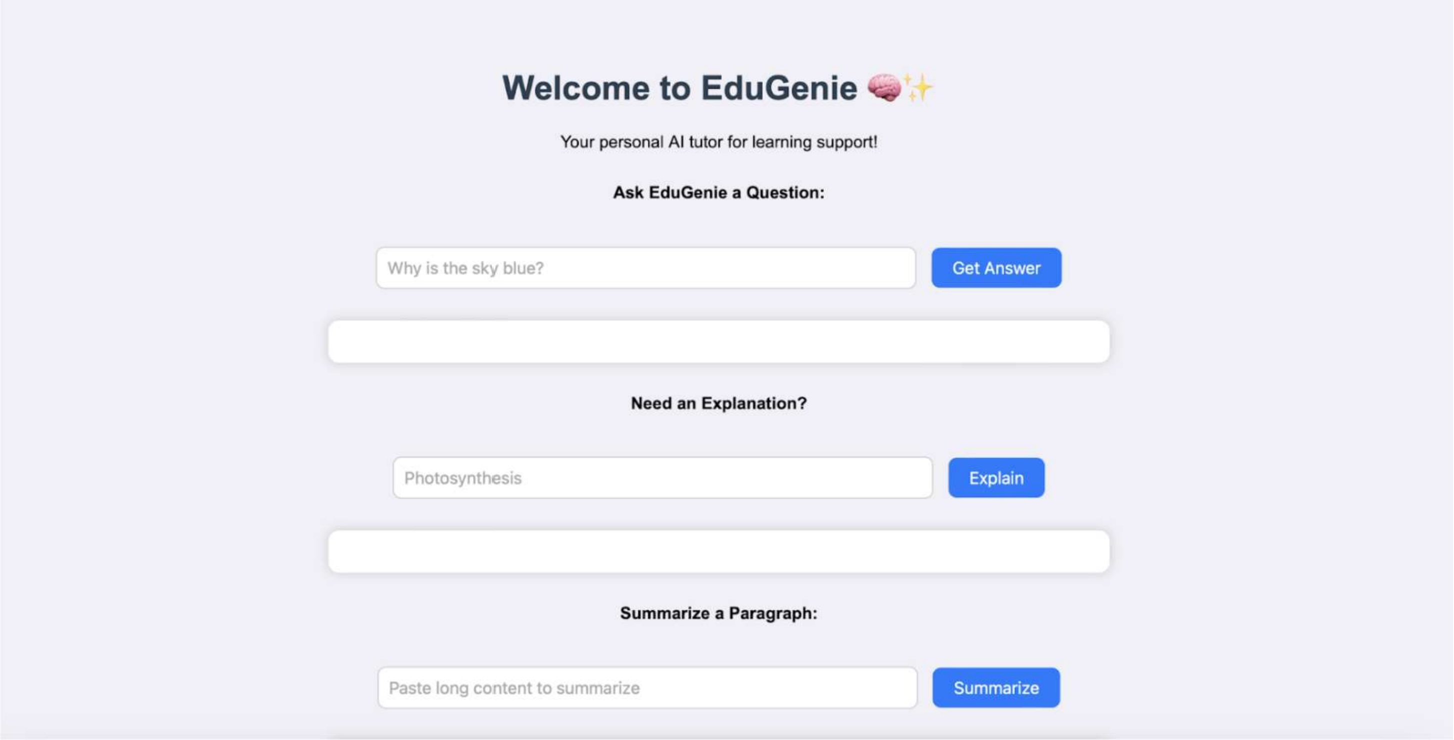
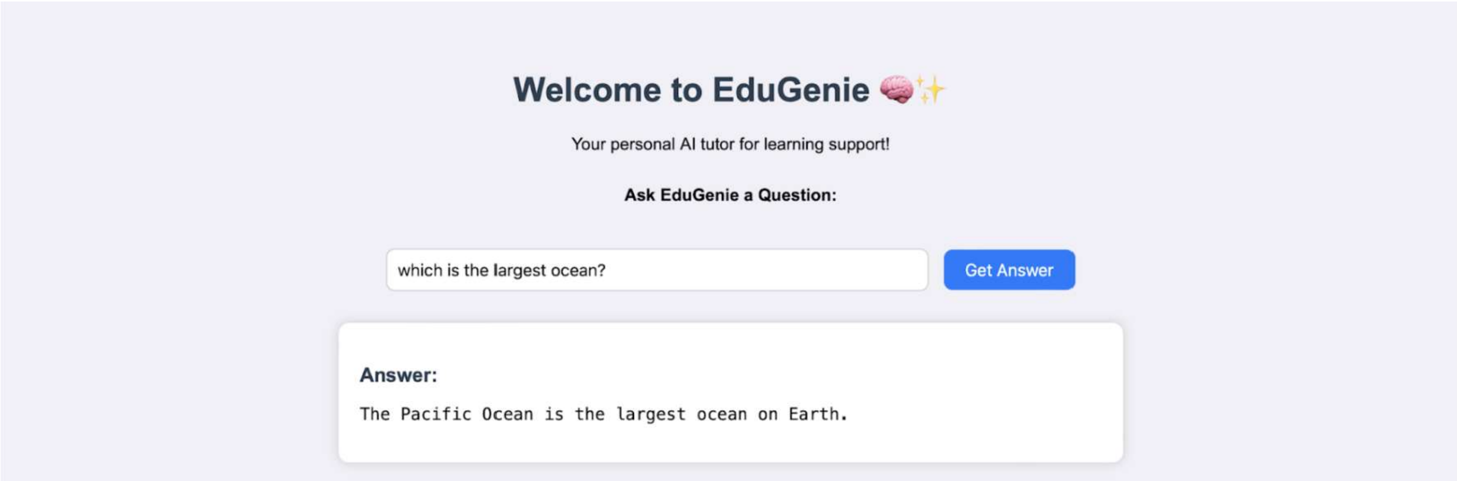
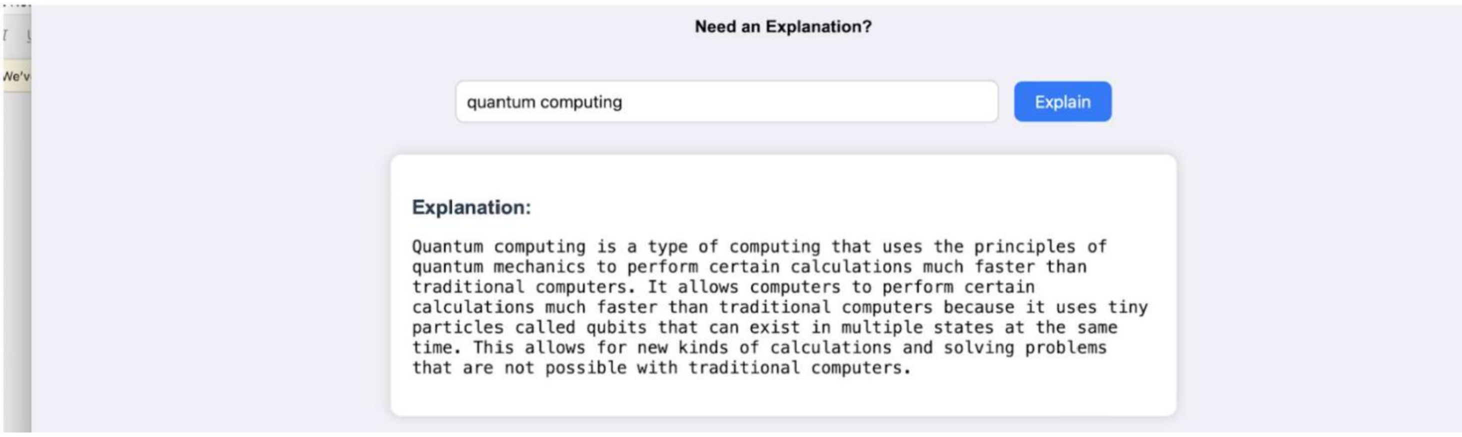


Fig. EDUGENIE

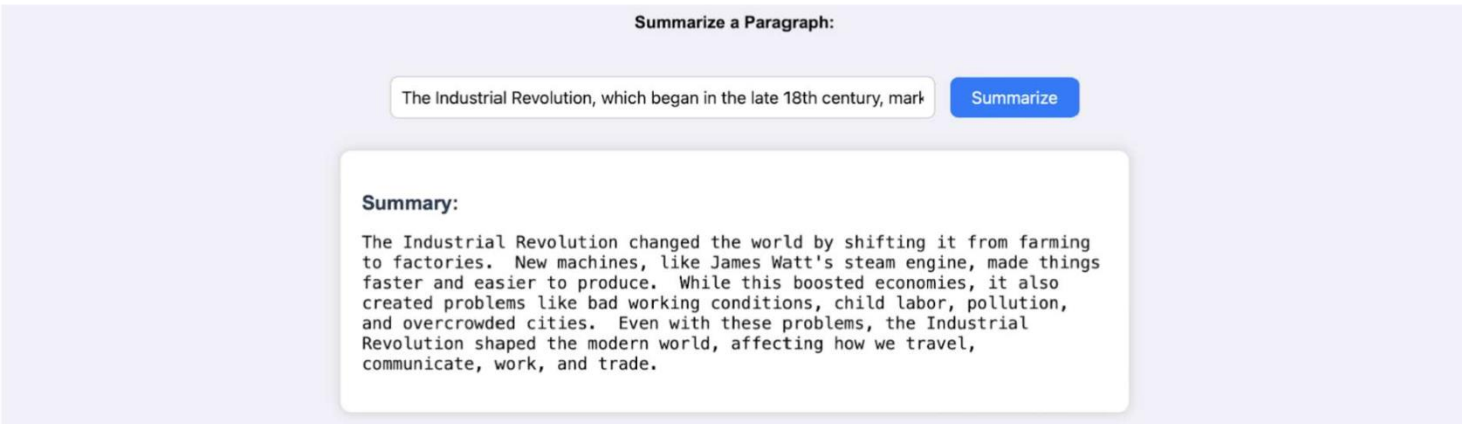
HomePage



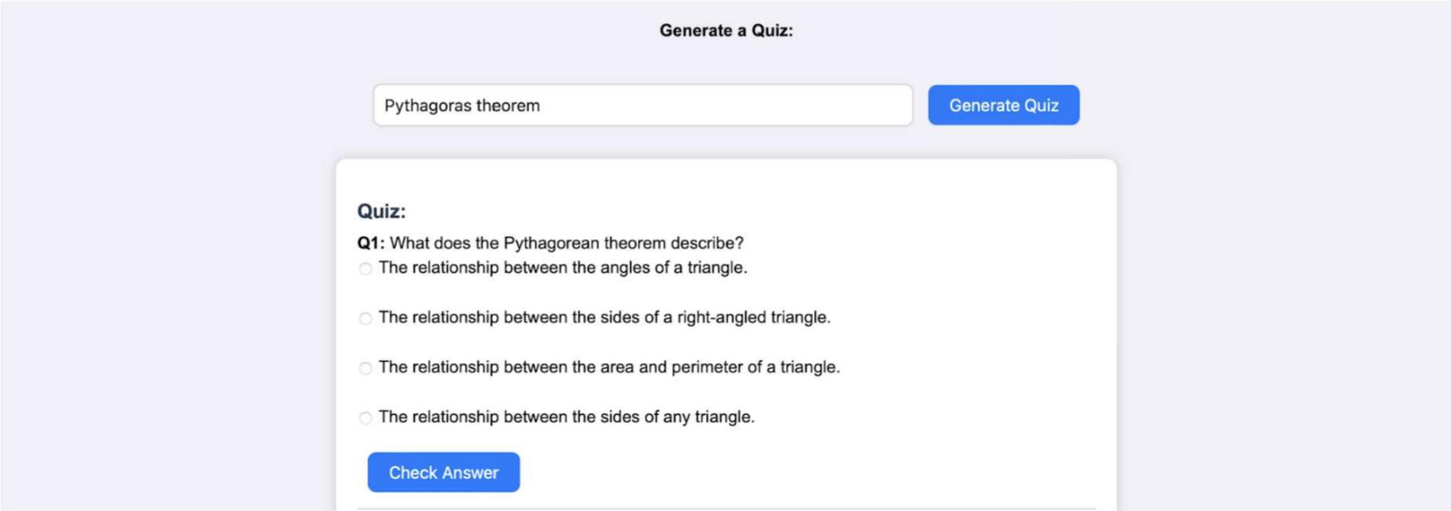
QuestionAnswering



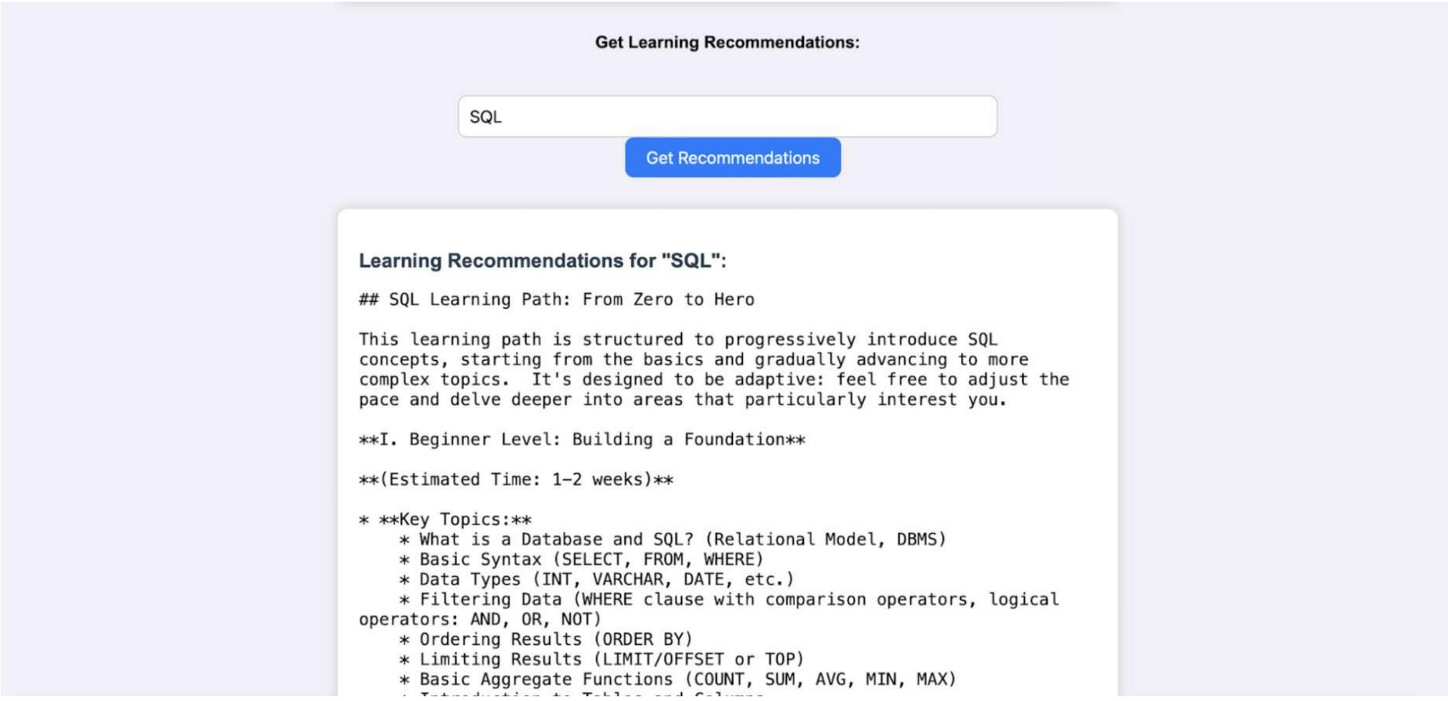
ExplanationOutput



EducationalTextSummary



AI-GeneratedQuiz



PersonalizedLearningPathRecommendation